

Formula/ Constant Sheet
Formule- en konstante bladsy

Circle: Circumference $= 2\pi r =$
omtrek

$$\text{Area} = \pi r^2$$

Sphere: Area $= 4\pi r^2$

$$\text{Volume} = \frac{4}{3}\pi r^3$$

Cylinder / Silinder

$$\text{Area} = 2(\pi r^2) + 2\pi rh$$

$$\text{Volume} = \pi r^2 h$$

Trapesium: Area $= \frac{1}{2}(a+b)h$

$$\vec{a} \cdot \vec{b} = ab \cos \phi$$

$$\vec{a} \cdot \vec{b} = a_x b_x + a_y b_y + a_z b_z$$

$$\vec{a} \times \vec{b} = (a_y b_z - b_y a_z) \hat{i}$$

$$+ (a_z b_x - b_z a_x) \hat{j}$$

$$+ (a_x b_y - b_x a_y) \hat{k}$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$v_{avg} = \frac{1}{2}(v_0 + v)$$

$$v = v_0 + at$$

$$x = x_0 + v_0 t + \frac{1}{2}at^2$$

$$x = x_0 + v_{avg} t$$

$$v^2 = v_0^2 + 2a(x - x_0)$$

$$x - x_0 = \frac{1}{2}(v_0 + v)t$$

$$x - x_0 = vt - \frac{1}{2}at^2$$

$$a = \frac{v^2}{r}$$

$$g = 9,8 \text{ m/s}^2$$

$$h = \frac{v_0^2 \sin^2 \theta}{2g}$$

$$R = \frac{v_0^2}{g} \sin 2\theta_0$$

$$T = \frac{2v_0 \sin \theta}{g}$$

$$f = \mu N$$

$$D = \frac{1}{2} C \rho A v^2$$

$$W = \vec{F} \cdot \vec{d} = Fd \cos \phi$$

$$F = -kx$$

$$W = -\frac{1}{2}kx^2$$

$$P_{avg} = \frac{W}{\Delta t}$$

$$P = \frac{dW}{dt}$$

$$P = \vec{F} \cdot \vec{v}$$

$$U = \frac{1}{2}kx^2$$

$$x_{com} = \frac{1}{M} \sum_{i=1}^n m_i x_i$$

$$\vec{r}_{com} = \frac{1}{M} \sum_{i=1}^n m_i \vec{r}_i$$

$$\vec{p} = m\vec{v}$$

$$\vec{F}_{net} = \frac{d\vec{p}}{dt}$$

$$\Delta \vec{p} = \vec{J}$$

$$\vec{J} = \int_{t_i}^{t_f} \vec{F}(t) dt$$

$$J = F_{avg} \Delta t$$

$$v_{1f} = \frac{m_1 - m_2}{m_1 + m_2} v_{1i}$$

$$v_{2f} = \frac{2m_1}{m_1 + m_2} v_{1i}$$

$$\theta = \frac{s}{r}$$

$$\omega_{avg} = \frac{\Delta \theta}{\Delta t}$$

$$\omega = \frac{d\theta}{dt}$$

$$\alpha_{avg} = \frac{\Delta \omega}{\Delta t}$$

$$\alpha = \frac{d\omega}{dt}$$

$$\omega = \omega_0 + \alpha t$$

$$\theta - \theta_0 = \omega_0 t + \frac{1}{2} \alpha t^2$$

$$\omega^2 = \omega_0^2 + 2\alpha(\theta - \theta_0)$$

$$\theta - \theta_0 = \frac{1}{2}(\omega + \omega_0)t$$

$$\theta - \theta_0 = \omega t - \frac{1}{2} \alpha t^2$$

$$s = \theta r$$

$$v = \omega r$$

$$a_t = \alpha r$$

$$a_r = \frac{v^2}{r} = \omega^2 r$$

$$T = \frac{2\pi r}{v} = \frac{2\pi}{\omega}$$

$$\tau = rF \sin \phi$$

$$\tau = I\alpha$$

$$I = \sum m_i r_i^2$$

$$I = \int r^2 dm$$

$$I = I_{com} + Mh^2$$

$$T = \frac{1}{f}$$

$$x = x_m \cos(\omega t + \phi)$$

$$\omega = \frac{2\pi}{T} = \sqrt{\frac{k}{m}}$$

$$T = 2\pi \sqrt{\frac{I}{mgh}}$$

$$T = 2\pi \sqrt{\frac{L}{g}}$$

$$y(x, t) = y_m \sin(kx - \omega t)$$

$$k = \frac{2\pi}{\lambda}$$

$$v = \frac{\omega}{k} = \lambda f$$

$$y'(x, t) = [2y_m \cos \frac{1}{2}\phi] \sin(kx - \omega t + \frac{1}{2}\phi)$$

$$y'(x, t) = [2y_m \sin kx] \cos(\omega t)$$

$$f = \frac{v}{\lambda} = \frac{nv}{2L}$$

$$\phi = \frac{\Delta L}{\lambda} 2\pi$$

$$\phi = m(2\pi), \quad m = 1, 2, 3, \dots$$

$$\phi = (m + \frac{1}{2})(2\pi), \quad m = 0, 1, 2, \dots$$

$$f = \frac{v}{\lambda} = \frac{nv}{2L}, \quad n = 1, 2, 3, \dots$$

$$f = \frac{v}{\lambda} = \frac{nv}{4L}, \quad n = 1, 3, 5, \dots$$

$$f_{beat} = f_1 - f_2$$

$$f' = f \frac{v \pm v_D}{v \pm v_S}$$

$$T_c = T_K - 273.15^\circ$$

$$T_F = \frac{9}{5} T_C + 32$$

$$\Delta L = L\alpha \Delta T$$

$$\Delta V = V\beta \Delta T$$

$$Q = C(T_f - T_i)$$

$$Q = cm(T_f - T_i)$$

$$Q = Lm$$

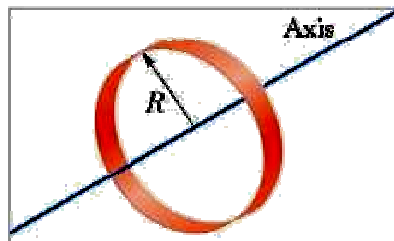
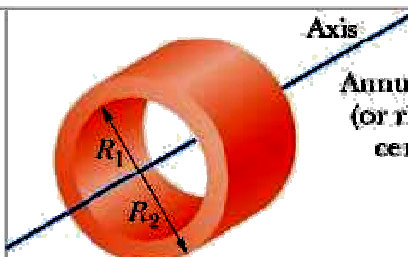
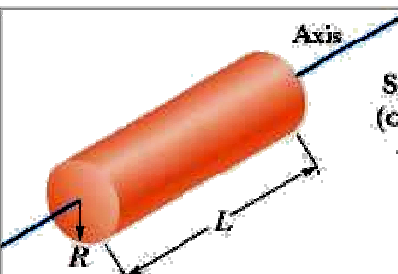
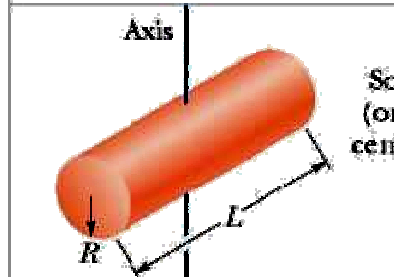
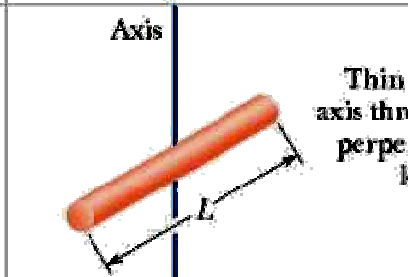
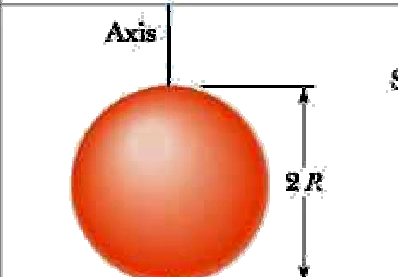
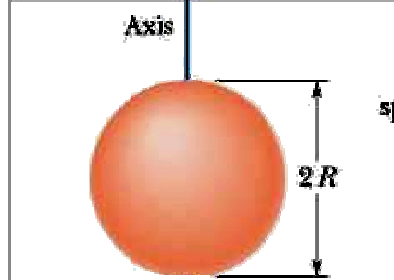
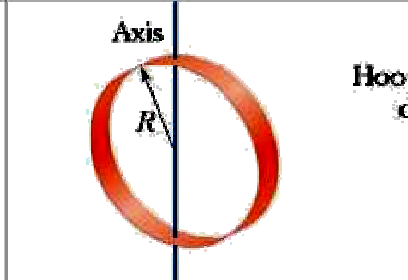
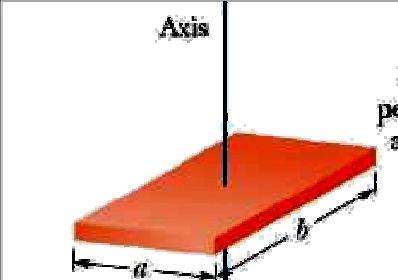
$$W = \int_{v_i}^{v_f} p dV$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

$$\sin(\alpha \pm \beta) = \sin \alpha \cos \beta \pm \cos \alpha \sin \beta$$

$$\cos(\alpha \pm \beta) = \cos \alpha \cos \beta \mp \sin \alpha \sin \beta$$

 Hoop about central axis $I = MR^2$ (a)	 Annular cylinder (or ring) about central axis $I = \frac{1}{2}M(R_1^2 + R_2^2)$ (b)	 Solid cylinder (or disk) about central axis $I = \frac{1}{2}MR^2$ (c)
 Solid cylinder (or disk) about central diameter $I = \frac{1}{4}MR^2 + \frac{1}{12}ML^2$ (d)	 Thin rod about axis through center perpendicular to length $I = \frac{1}{12}ML^2$ (e)	 Solid sphere about any diameter $I = \frac{2}{5}MR^2$ (f)
 Thin spherical shell about any diameter $I = \frac{2}{3}MR^2$ (g)	 Hoop about any diameter $I = \frac{1}{2}MR^2$ (h)	 Slab about perpendicular axis through center $I = \frac{1}{12}M(a^2 + b^2)$ (i)